L1.8 Linearization & Taylor

Work in your groups. Answers are in the document you get at the end — don't wait for it.

Linearization

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1. Find the linearization of $f(x) = \dfrac{2}{\sqrt{x^2 - 5}}$ at $a = 3$.

2. Use it to estimate $f(3.1)$. Then say whether your estimate is too big or too small, and how you know **without** computing $f(3.1)$.

3. Estimate $(2.001)^5$ without a calculator.

4. A function is **concave down** everywhere. Sketch it with a tangent line drawn at some point a, and decide: is $L(x)$ an over-estimate or an under-estimate of $f(x)$? Does your answer depend on which side of a you are on?

Differentials and actual change

5. Find the differential dy.

(a) $y = \dfrac{1+2x}{1+3x}$

(b) $y = x^2 \sin(2x)$

6. For $y = x^3$ at $x = 2$ with $\Delta x = dx = 0.1$:

(a) Find Δy.

(b) Find dy.

(c) Which one did the tangent line give you, and what is the difference between them?

Error propagation

Stop. An angle is about to be multiplied by a length. Which measure does the angle have to be in — and what is the reason, not just the rule?

7. One side of a right triangle is known to be 20 cm long, and the angle opposite it is measured as $30°$, with a possible error of $1°$.

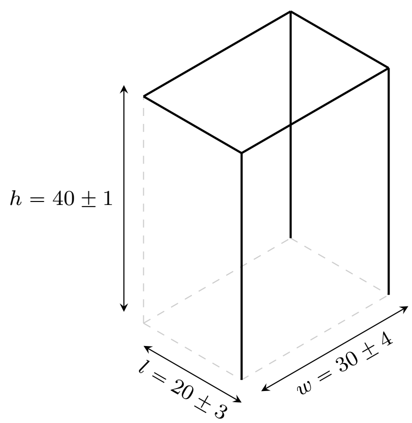
(a) Use differentials to estimate the error in the computed length of the hypotenuse.

(b) What is the percentage error?

8. A sphere's radius is measured to within 2%. To within what percentage is its volume known? Its surface area?

Stop. You know what a fractional uncertainty does to r^3 — one length, raised to a power. The box below is $l\,w\,h$: three *different* lengths, each with its own uncertainty. What do you think happens?

9. The length, width and height of a box are each known only to some uncertainty: $(l, w, h) = (20 \pm 3\text{ cm},\ 30 \pm 4\text{ cm},\ 40 \pm 1\text{ cm})$.



(a) What is the fractional uncertainty of each dimension?

(b) What is the fractional uncertainty of the volume?

(c) What is the uncertainty of the volume?

Summation notation and the closed forms

10. Evaluate.

(a) $\displaystyle\sum_{i=1}^{4} \left(i^2 + 1\right)$ — write out every term.

(b) $\displaystyle\sum_{i=1}^{30} i^3$ — use a closed form.

(c) Check that $\displaystyle\sum_{i=1}^{4} i^3$ really does equal $\displaystyle\left(\sum_{i=1}^{4} i\right)^2$.

Maclaurin and Taylor series

11. Maclaurin series for $f(x) = \dfrac{1}{2-x}$.

n	$f^{(n)}(x)$	$f^{(n)}(0)$	$\dfrac{f^{(n)}(0)}{n!}$
0			
1			
2			
3			

$T_3(x) =$ ____

12. Maclaurin series for $f(x) = \dfrac{1}{1+x^2}$.

n	$f^{(n)}(x)$	$f^{(n)}(0)$	$\dfrac{f^{(n)}(0)}{n!}$
0			
1			
2			
3			

$T_3(x) =$ ____

13. Taylor series for $f(x) = \sqrt{x}$, centred at $a = 4$.

n	$f^{(n)}(x)$	$f^{(n)}(4)$	$\dfrac{f^{(n)}(4)}{n!}$
0			
1			
2			
3			

$T_3(x) =$ ____

Powers of $(x - 4)$, not powers of x.

14. BOARDS Estimate $\sqrt{10}$ without a calculator. Get as many decimal places as you can **defend** — be ready to say why you trust each one.